

Shubham Sarkar

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Education

Jadavpur University, Kolkata <i>Bachelor of Technology</i>	<i>Nov 2023 – Dec 2027</i> <i>CGPA: 7.5</i>
Hariyana Vidya Mandir, Kolkata <i>Higher Secondary</i>	<i>Apr 2020 – Apr 2022</i> <i>Percentage: 90%</i>

Experience

Alignerr <i>Machine Learning Engineer</i>	<i>Jan 2026 – Present</i> <i>Remote</i>
- Enhanced LLM evaluation precision by 15% through a comprehensive review of a rubric-based scoring framework across 6 reasoning categories. - Analyzed over 50 audio files for integration into ASR pipelines. - Evaluated AI agent responses, identifying failure points such as Inference memory and Self Coherence, leading to a 20% improvement in model accuracy.	
Jadavpur University CMATER Lab <i>Deep Learning Research Assistant</i>	<i>May 2025 – September 2025</i> <i>Kolkata, India</i>
- Designed and implemented a self-attention mechanism (scaled dot-product) within a pre-trained VGG16, enhancing feature extraction for lung cancer detection from CT scans. - Developed a hybrid deep learning architecture achieving 99.54% peak accuracy with only 76k trainable parameters and 0.0256 GFLOPs, enabling edge-device deployment. - Engineered feature fusion via concatenation and element-wise multiplication of original and attention-modulated maps for refined, context-aware representations.	

Projects

TailorCV.ai <i>Python, FastAPI, LLM, AI agents, Amazon Web Services</i> GitHub ↗ Live Demo ↗	<i>2 January 2026</i>
- Developed an AI web application that optimizes resumes to job descriptions using LLMs and NLP pipelines, improving resume relevance by up to 80%. - Designed a Python and FastAPI backend with HTML, CSS, and JavaScript for the frontend, dockerized it, and deployed on AWS ECS. - Achieved over 50 users within the first week of launch, demonstrating strong early adoption and real-world impact.	
YouTube Sentiment Analysis <i>TensorFlow, NLP, AWS EC2, Scikit-learn</i>	<i>2 October 2025</i>
- Created an end-to-end YouTube sentiment analysis pipeline processing over 10k user comments, enhancing sentiment classification performance through NLP preprocessing techniques. - Tracked multiple model experiments using MLflow and DVC, enabling reproducible training and systematic comparison of models built with scikit-learn and NLP libraries. - Deployed the pipeline on AWS using Docker and exposed predictions via Flask REST APIs, facilitating scalable and reproducible inference.	
Smart Product Pricing <i>Keras, Hugging Face Transformers, ResNet50, OpenCV</i> Kaggle ↗	<i>2 October 2025</i>
- Developed an NLP and CV pipeline to analyze 150,000 image and text data using transformer-based text encoders and CNN-based image embeddings, integrating them through a fusion neural network for price prediction. - Implemented data preprocessing, including text cleaning, tokenization, and streaming image feature extraction with ResNet and CLIP representations to handle large datasets. - Built and fine-tuned models using TensorFlow and scikit-learn, achieving a rank of 142 out of 50,000 participants.	

Technical Skills

Languages: Python, SQL
AI/ML: Machine Learning, MLOps
Frameworks/Libraries: Pandas, NumPy, Scikit-learn, Statsmodels, Scipy, Spark, TensorFlow, PyTorch, Hugging Face, Flask, FastAPI, ETL Pipelines
Databases: Databricks, PostgreSQL, Vector Databases
Tools & Platforms: MLflow, DVC
Cloud & DevOps: Azure, AWS, Docker, Linux
Other Technical Skills: Statistical Modelling, Forecasting, Time-Series Analysis, Model Explainability, Hypothesis Testing, Predictive Analytics, Prophet, Geospatial Analytics, GeoPandas, Shapely, time-series forecasting, model interpretation, geospatial data analysis, modelling

Leadership / Extracurricular

Core Member at Entrepreneurship Cell, Jadavpur University

May 2024 – Present

- Organized national-level events such as E-Summit 2025 and Hult Prize 2025, attracting over 5,000 registrations and 1,000+ attendees.
- Contributed to the establishment of an Incubation Center at Jadavpur University under the Institution’s Innovation Council (IIC).